

A Corrected Progress Note on the $1 - 1/e$ Question for Online Bipartite Matching with Reusable Resources

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Abstract

This note revisits an attempted proof of a $(1 - 1/e)$ -competitive integral algorithm for online bipartite matching with unit-inventory deterministic reusable resources. I do *not* obtain a solution of the open problem of Delong, Farhadi, Niazadeh, Sivan, and Udwani. The contribution here is partial and corrective: I isolate which parts of the attempted proof are valid, prove several boundary and decomposable cases, give a self-contained obstruction to black-box lossless rounding from the natural LP, and prove a two-arrival witness showing why the classical one-seed monotonicity argument does not transfer to reranking-on-return. The main open problem, namely an integral $(1 - 1/e)$ guarantee for arbitrary OBMRR instances, remains open after this note.

Status. The result below is **partial progress only**. It is not a proof of a $(1 - 1/e)$ -competitive algorithm for arbitrary OBMRR. The previous draft’s proposed “renewal-threshold” argument is not repaired here; instead, the note records rigorous statements that survive verification and explains why that threshold route cannot be justified by the usual Ranking monotonicity alone.

1 Model and the target question

An instance consists of a bipartite graph

$$G = (V, U; E), \quad U = \{1, 2, \dots, T\},$$

where resources $i \in V$ are known in advance and arrivals $j \in U$ appear online in increasing time order. Resource i has nonnegative reward r_i . When arrival j appears, its incident edges to resources are revealed. The algorithm must either leave j unmatched or irrevocably match it to an available neighboring resource. If resource i is matched at time j , then it is unavailable at times $j + 1, \dots, j + d - 1$ and becomes available again at time $j + d$. Thus each resource can be used at most once in every length- d time window.

The reward of an integral schedule M is

$$w(M) = \sum_{(i,j) \in M} r_i.$$

A randomized online algorithm is Γ -competitive if, for every adversarially chosen input instance,

$$\mathbb{E}[w(M_{\text{ALG}})] \geq \Gamma w(M_{\text{OPT}}),$$

where M_{OPT} is the optimal offline integral reusable matching.

Delong et al. [3] gave the first algorithms beating the greedy $1/2$ barrier for this unit-inventory model. Their best stated guarantee is approximately 0.589 via Periodic Reranking. They explicitly leave open whether the classical $(1 - 1/e)$ ratio can be achieved for OBMRR. Since the nonreusable online bipartite matching problem is recovered when $d \geq T$, the best possible universal ratio for the whole family of OBMRR instances is at most $1 - 1/e$.

2 The natural LP and a dual-fitting certificate

The standard relaxation has variables x_{ij} for $(i, j) \in E$:

$$\max \sum_{(i,j) \in E} r_i x_{ij} \tag{P}$$

$$\text{s.t.} \quad \sum_{i:(i,j) \in E} x_{ij} \leq 1 \quad \forall j \in U, \tag{P1}$$

$$\sum_{\tau=t}^{t+d-1} x_{i\tau} \leq 1 \quad \forall i \in V, t = 1, \dots, T, \tag{P2}$$

$$x_{ij} \geq 0 \quad \forall (i, j) \in E.$$

Here $x_{i\tau} = 0$ when $(i, \tau) \notin E$ or $\tau \notin [T]$. The dual is

$$\min \sum_{j \in U} \beta_j + \sum_{i \in V} \sum_{t=1}^T \alpha_{i,t} \tag{D}$$

$$\begin{aligned} \text{s.t.} \quad & \beta_j + \sum_{t=\max\{1, j-d+1\}}^j \alpha_{i,t} \geq r_i \quad \forall (i, j) \in E, \\ & \alpha_{i,t}, \beta_j \geq 0. \end{aligned} \tag{D1}$$

Lemma 1 (Dual fitting). *Suppose an online algorithm induces a feasible primal vector x whose objective value is $P = \mathbb{E}[w(M_{\text{ALG}})]$. Suppose there are nonnegative dual variables (α, β) with objective value*

$$D = \sum_j \beta_j + \sum_{i,t} \alpha_{i,t}$$

such that

$$P \geq D$$

and, for some $\Gamma \in [0, 1]$,

$$\beta_j + \sum_{t=\max\{1, j-d+1\}}^j \alpha_{i,t} \geq \Gamma r_i \quad \forall (i, j) \in E.$$

Then the algorithm is Γ -competitive against the integral offline optimum.

Proof. If $\Gamma = 0$ there is nothing to prove. For $\Gamma > 0$, the scaled variables $(\alpha/\Gamma, \beta/\Gamma)$ are feasible for the dual (D). By weak duality,

$$\frac{D}{\Gamma} \geq \text{LP} \geq w(M_{\text{OPT}}),$$

where LP is the value of (P). Hence $P \geq D \geq \Gamma w(M_{\text{OPT}})$. $\square$

3 Verified baseline guarantees

Proposition 2 (Universal upper bound). *For the family of all deterministic-duration unit-inventory OBMRR instances,*

$$\sup_{\text{ALG}} \inf_I \frac{\mathbb{E}[\text{ALG}(I)]}{\text{OPT}(I)} \leq 1 - \frac{1}{e}.$$

Proof. Take $d \geq T$. Then a resource, once matched, never returns during the horizon. The model contains classical online bipartite matching, and the Karp–Vazirani–Vazirani lower bound rules out a randomized competitive ratio larger than $1 - 1/e$ even in the unweighted special case. Therefore no algorithm for the larger OBMRR family can have a better universal guarantee. $\square$

Remark 3. This upper bound is global in d . For a fixed small duration, for example $d = 2$, this reduction does not imply any $1 - 1/e$ upper bound.

Proposition 4 (Weighted greedy). *The deterministic algorithm that matches each arrival to an available neighbor of maximum reward is $1/2$ -competitive for vertex-weighted OBMRR.*

Proof. Fix the input and compare the greedy matching M with an optimal matching O . Charge each optimum edge $e = (i, j) \in O$ to a greedy edge.

If resource i is available to greedy at time j , then greedy either matches j to some i' with $r_{i'} \geq r_i$ or leaves j unmatched only when every available neighbor has reward 0; in either case, if $r_i > 0$, charge e to the greedy edge incident to j . If $r_i = 0$, the charge is harmless. If i is not available to greedy at time j , then greedy matched i at some time

$$t \in \{j - d + 1, j - d + 2, \dots, j - 1\},$$

and we charge e to that greedy edge (i, t) .

A greedy edge (i, t) receives at most one same-arrival charge, because O matches arrival t at most once. It also receives at most one blocking charge from optimum edges using resource i in times $t + 1, \dots, t + d - 1$, because O itself can use i at most once in the length- d window $t, t + 1, \dots, t + d - 1$. Each charge has weight at most the reward of the greedy edge to which it is charged. Hence

$$w(O) \leq 2w(M).$$

$\square$

Proposition 5 (One resource). *If $|V| = 1$, the earliest-available greedy algorithm is optimal for every d .*

Proof. Let the single resource be i , and let eligible arrival times be those adjacent to i . Greedy chooses the first eligible time g_1 , then the first eligible time $g_2 \geq g_1 + d$, and so on. Let $o_1 < o_2 < \dots$ be the times chosen by any feasible offline solution. We prove by induction that $g_k \leq o_k$ whenever both sides exist. The claim is clear for $k = 1$, since g_1 is the earliest eligible time. If $g_k \leq o_k$, then $o_{k+1} \geq o_k + d \geq g_k + d$, so o_{k+1} is a feasible candidate for greedy's next choice after g_k . Therefore $g_{k+1} \leq o_{k+1}$. Thus greedy makes at least as many matches as any offline solution. $\square$

Proposition 6 (Duration one). *If $d = 1$, the maximum-reward greedy algorithm is optimal.*

Proof. A match at time j causes no unavailability at any later arrival. Hence the decision at each arrival is independent of all other decisions. The offline optimum and the online greedy algorithm both obtain, at each arrival j , the largest reward among neighbors of j , or zero if j has no neighbor. $\square$

4 A restricted positive result: cooling-separated blocks

The following theorem is a genuine $(1 - 1/e)$ result, but only for a strongly restricted subclass. It is included because it is the cleanest way to repair the classical Ranking/Perturbed-Greedy proof: one must remove cross-block reusable-resource dependence.

Definition 7 (Cooling-separated block instance). An OBMRR instance is cooling-separated if the nonempty arrivals can be partitioned into consecutive blocks

$$B_m = \{s_m, s_m + 1, \dots, t_m\}$$

such that

$$t_m - s_m + 1 \leq d \quad \text{and} \quad s_{m+1} \geq t_m + d$$

for every m . Arrivals outside the blocks have no incident edges.

Theorem 8 (Perturbed greedy on cooling-separated blocks). *On cooling-separated block instances, running the classical vertex-weighted Perturbed Greedy algorithm independently in each block is $(1 - 1/e)$ -competitive. In fact, its expected reward is at least $(1 - 1/e)$ times the offline optimum on this restricted OBMRR instance.*

Proof. Inside a block B_m , any two arrival times differ by at most $d - 1$. Therefore no feasible schedule can use the same resource twice inside that block. Thus the restriction of the problem to one block is exactly a classical nonreusable vertex-weighted online bipartite matching instance.

The separation condition $s_{m+1} \geq t_m + d$ implies that any resource matched at any time in B_m has returned by the first arrival of B_{m+1} . Hence feasibility constraints do not couple different blocks. Consequently the offline optimum decomposes additively:

$$\text{OPT} = \sum_m \text{OPT}(B_m).$$

Run Perturbed Greedy with fresh independent perturbations at the beginning of each block. The Aggarwal–Goel–Karande–Mehta theorem gives expected reward at least $(1 - 1/e) \text{OPT}(B_m)$ in every block. Summing over blocks gives the claim. $\square$

Remark 9. The theorem is intentionally limited. Arbitrary OBMRR instances do not have cooling gaps, and the hard part of the open problem is precisely to handle the dependence created by matches that affect nearby future arrivals while random ranks may also need to decorrelate far-away choices.

5 Why black-box lossless rounding is not a repair

A previous attempted route was: run a fractional $(1 - 1/e)$ -competitive balance algorithm, then round it online while preserving every edge marginal. As a conditional statement this is true, but the required rounding is the hard part. Moreover, for the natural OBMRR LP, a black-box lossless rounding of every feasible fractional point is impossible even offline.

Definition 10 (Lossless marginal rounding). Given a feasible fractional vector x for $(\mathbf{P})$, a lossless marginal rounding is a distribution over feasible integral reusable matchings M such that

$$\mathbb{P}[(i, j) \in M] = x_{ij} \quad \forall (i, j) \in E.$$

Proposition 11 (No lossless rounding for all LP points). *There are feasible points of the natural OBMRR LP that admit no lossless marginal rounding, even with full offline knowledge of the instance.*

Proof. Consider the unweighted instance with $d = 3$, resources $V = \{v_1, v_2, v_3\}$, arrivals $U = \{u_1, u_2, u_3, u_4\}$, and edges

$$E = \{(v_1, u_1), (v_1, u_2), (v_2, u_2), (v_2, u_4), \\ (v_3, u_1), (v_3, u_3), (v_3, u_4)\}.$$

Set $x_e = 1/2$ for every edge $e \in E$. Each arrival is incident to at most two of these fractional half-edges, so every arrival constraint is satisfied. For v_1 , the only incident times are 1, 2 and the total in every length-3 window is at most 1. For v_2 , the incident times are 2, 4, and the only length-3 window containing both has total 1. For v_3 , the incident times are 1, 3, 4; the windows $\{1, 2, 3\}$ and $\{2, 3, 4\}$ each contain total fractional mass 1, and no window contains more. Thus x is feasible, with value

$$\sum_{e \in E} x_e = \frac{7}{2}.$$

Every integral feasible matching has size at most 3. Indeed, suppose all four arrivals were matched. Arrival u_3 must use v_3 . Then u_1 cannot use v_3 and must use v_1 . Since $d = 3$, resource v_1 is unavailable at u_2 , so u_2 must use v_2 . But then v_2 is unavailable at u_4 , and v_3 is also unavailable at u_4 because it was used at u_3 . This contradiction shows $\text{OPT} = 3$.

If a lossless marginal rounding of x existed, the expected size of the rounded integral matching would be

$$\mathbb{E}[|M|] = \sum_{e \in E} \mathbb{P}[e \in M] = \sum_{e \in E} x_e = \frac{7}{2},$$

which is impossible because every feasible integral matching has size at most 3. $\square$

Corollary 12. *A rounding-based proof of a $(1 - 1/e)$ integral guarantee cannot be a black-box lossless rounding theorem for all feasible points of the natural OBMRR LP. It would have to exploit additional structure of a particular fractional algorithm's trajectory or use a different relaxation.*

Proof. This is immediate from Proposition 11. $\square$

6 Reranking-on-return and the failed monotonicity template

Definition 13 (Reranking-on-return). In RoR, every available resource i has an independent rank $Y_i \sim \text{Unif}[0, 1]$. While i remains available, this rank is kept fixed. If i is matched, it is removed for $d - 1$ arrivals. When it returns, it receives a fresh independent rank. On arrival j , the algorithm matches j to the available neighbor maximizing

$$r_i(1 - e^{Y_i - 1}),$$

with deterministic tie-breaking.

For $d \geq T$, RoR is the classical Perturbed Greedy algorithm and is $(1 - 1/e)$ -competitive. For general d , the main difficulty is that changing the rank of a resource can change not merely whether it wins a particular arrival, but also whether it is consumed earlier and hence which future rank it has when it returns.

Proposition 14 (A minimal anti-monotonicity witness). *There is a two-arrival instance with $d = 2$ on which the event that a fixed optimum edge is chosen by RoR is monotone in the opposite direction from the classical Ranking threshold event.*

Proof. Let $V = \{a, b\}$, let both rewards be 1, and let $d = 2$. Arrival 1 is adjacent to both resources, and arrival 2 is adjacent only to a .

Fix the rank $Y_b = c \in (0, 1)$ of resource b and vary the initial rank $Y_a = y$ of resource a . Since lower rank gives larger score, RoR matches arrival 1 to a exactly when $y < c$. In that case a is unavailable at arrival 2, so edge $(a, 2)$ is not chosen. If $y > c$, arrival 1 is matched to b , and a remains available for arrival 2, so edge $(a, 2)$ is chosen. Therefore

$$\mathbf{1}\{(a, 2) \text{ is chosen by RoR}\} = \mathbf{1}\{Y_a > c\}.$$

This is increasing in Y_a , whereas the classical Ranking/Perturbed-Greedy threshold proof relies on better ranks, i.e. smaller Y_a , making the relevant resource-side event more favorable. $\square$

Proposition 15 (Exact value on the witness). *On the instance in Proposition 14, RoR has expected reward $3/2$, while the offline optimum has reward 2. Thus the ratio on this witness is $3/4$.*

Proof. With probability $1/2$, $Y_a < Y_b$, so RoR matches a at arrival 1 and then cannot match arrival 2; its reward is 1. With probability $1/2$, $Y_b < Y_a$, so RoR matches b at arrival 1 and a at arrival 2; its reward is 2. Hence

$$\mathbb{E}[\text{RoR}] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 = \frac{3}{2}.$$

The offline optimum matches b to arrival 1 and a to arrival 2, obtaining 2. $\square$

Remark 16. The ratio $3/4$ is above $1 - 1/e$, so Proposition 15 is not a lower bound against the desired competitive ratio. Its purpose is different: it rigorously shows that the ordinary one-seed threshold monotonicity used in the nonreusable proof does not survive in reusable systems. A successful analysis of RoR would need a more global invariant that accounts for early consumption, return times, and fresh ranks.

7 A precise conditional statement that remains true

The following statement is logically correct but should not be mistaken for a solution.

Theorem 17 (Conditional rounding reduction). *Suppose a fractional online algorithm for OBMR is ρ -competitive against the integral offline optimum, and suppose that its fractional trajectory admits an online rounding scheme which, for every edge (i, j) , outputs a feasible integral reusable matching M satisfying*

$$\mathbb{P}[(i, j) \in M] = x_{ij}.$$

Then there is a randomized integral online algorithm with competitive ratio ρ .

Proof. Run the fractional algorithm and the assumed rounding scheme simultaneously. The rounded matching is feasible by assumption. Its expected reward is

$$\mathbb{E}[w(M)] = \sum_{(i,j) \in E} r_i \mathbb{P}[(i, j) \in M] = \sum_{(i,j) \in E} r_i x_{ij},$$

which is the fractional algorithm's reward. By the assumed fractional guarantee this is at least ρ OPT. $\square$

Remark 18. The theorem only moves the difficulty into the rounding assumption. Proposition 11 shows that this assumption cannot hold for arbitrary feasible points of the natural LP. It might still hold for a specially constrained fractional trajectory, but no such construction is provided here.

8 Conclusion

The attempted $(1 - 1/e)$ proof is not repairable by simply filling in the proposed renewal-threshold lemma. The reusable setting breaks the one-rank monotonicity underlying the classical proof, and the natural LP cannot be rounded losslessly in a black-box way. The verified results in this note are:

- (i) the standard global $1 - 1/e$ upper bound;
- (ii) a self-contained dual-fitting lemma and the weighted greedy $1/2$ proof;
- (iii) optimality in the one-resource and $d = 1$ boundary cases;
- (iv) a $(1 - 1/e)$ guarantee for cooling-separated block instances;
- (v) a concrete LP obstruction to universal lossless marginal rounding;
- (vi) a minimal anti-monotonicity witness for RoR.

These constitute partial progress and cleanup, not a solution of the main open problem for arbitrary OBMRR.

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